

IAM Power User

[Edition 10]

[Last Update 240102]

Scenario

In a modern cloud operations center, an expert in cloud security is setting up a new IAM Power User account. This task is crucial for managing who can access what in their AWS environment. After creating the account, they log in to check if everything works correctly. They test various AWS services to make sure the new account has the right permissions. IAM Power User helps keep their cloud system secure and running smoothly.

Description

In this Lab we will first, create an IAM Power User account. Next, log in using the IAM Power User credentials. Then, test your access to IAM to ensure proper permissions and functionality. Finally, verify access to other AWS services to confirm broad capabilities and correct setup.

For any issues/help contact in Training WhatsApp Group or Support Ticketing System

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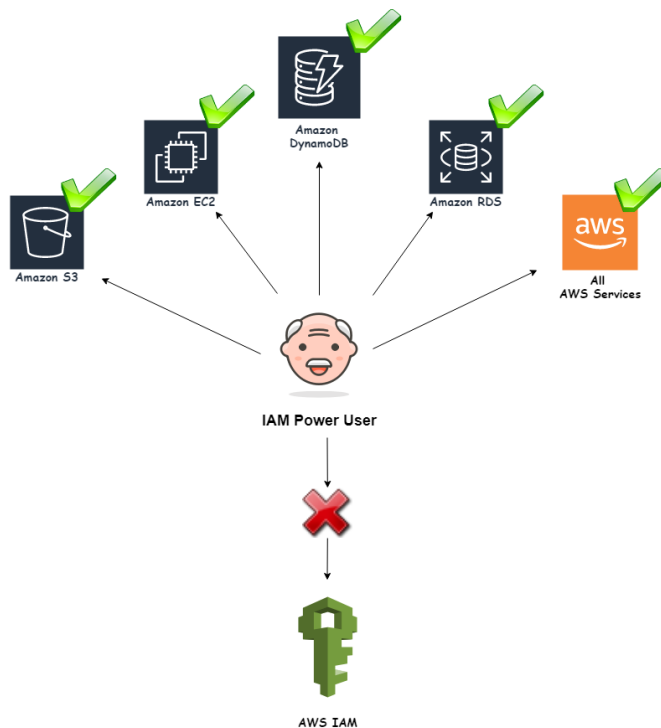
1 INTRODUCTION

The description of power user access given by AWS is “Provides full access to AWS services and resources, but does not allow management of Users and groups.”

The power to manage user is the highest privilege operation in AWS thus it is provided to the administrative access policy only. Power user is just below the admin access, and have all the privileges, other than the ability to manage IAM users.

The reason why ability to manage users is the highest privilege is simple. Suppose someone has been given full access to the IAM (Identity and Access Management) service only and no access to any other service but as the user is having the ability to manage user, she/he can give herself / himself access to any AWS services she/he wishes.

Architectural Diagram:



This activity guide covers steps for:

1. Creating an IAM Power User
2. Login the IAM Power user to AWS Console
3. Test IAM Service Access
4. Test other Service Access

2 OVERVIEW & KEY CONCEPTS

In this section we'll cover key concepts used in this guide, including an overview of what's covered in this guide.

2.1 What is IAM?

AWS Identity and Access Management (IAM) is a web service that allows you to regulate access to AWS services in a secure manner. IAM allows you to centrally manage permissions that govern which AWS resources users can access. IAM manages who has been authenticated (signed in) and authorized (has access to resources).

Whenever you start an AWS account, you start with a single sign-in identity with full access to all the account's AWS services and resources. This identity is known as the AWS account root user, and it may be accessed by logging in using the email address and password you used to establish the account. We highly advise against using the root user for routine operations.

2.2 What is a Power User?

You can use the AWS Key Management Service Power User managed policy to provide IAM principals in your account power user permissions. Power users have the ability to generate KMS keys, use and manage the keys they generate, and access all KMS keys and IAM identities. Principals with the AWS Key Management Service Power User managed policy can obtain permissions from a variety of sources, including key policies, other IAM policies, and grants.

Documentation: [What is IAM?](#)

Blog: <https://k21academy.com/amazon-web-services/aws-solutions-architect/aws-identity-and-access-management-iam/>

3 DOCUMENTATION LINKS

1. IAM Best Practices

<https://docs.aws.amazon.com/IAM/latest/UserGuide/best-practices.html>

2. AWS Managed policies and inline policies

https://docs.aws.amazon.com/IAM/latest/UserGuide/access_policies_managed-vs-inline.html

3. Example policies for administering IAM resources

https://docs.aws.amazon.com/IAM/latest/UserGuide/id_credentials_delegate-permissions_examples.html

4. AWS IAM FAQ's

<https://aws.amazon.com/iam/faqs/>

4 PRE-REQUISITES

An AWS Account (Free or Paid)

K21Academy

5 QUIZ QUESTION

Question: Which of the following AWS-defined IAM policies offer read and write access to the S3 and EC2 services? (Choose two.)

- A. Administrator
- B. Network Administrator
- C. Support User
- D. Power User

Answer: A, D.

Explanation: Both the Administrators and the Power Users default policies provide read and write access to most AWS services. Power Users limits access to IAM, but that would not affect access to S3 or EC2.

This Activity guide is based on this quiz question.

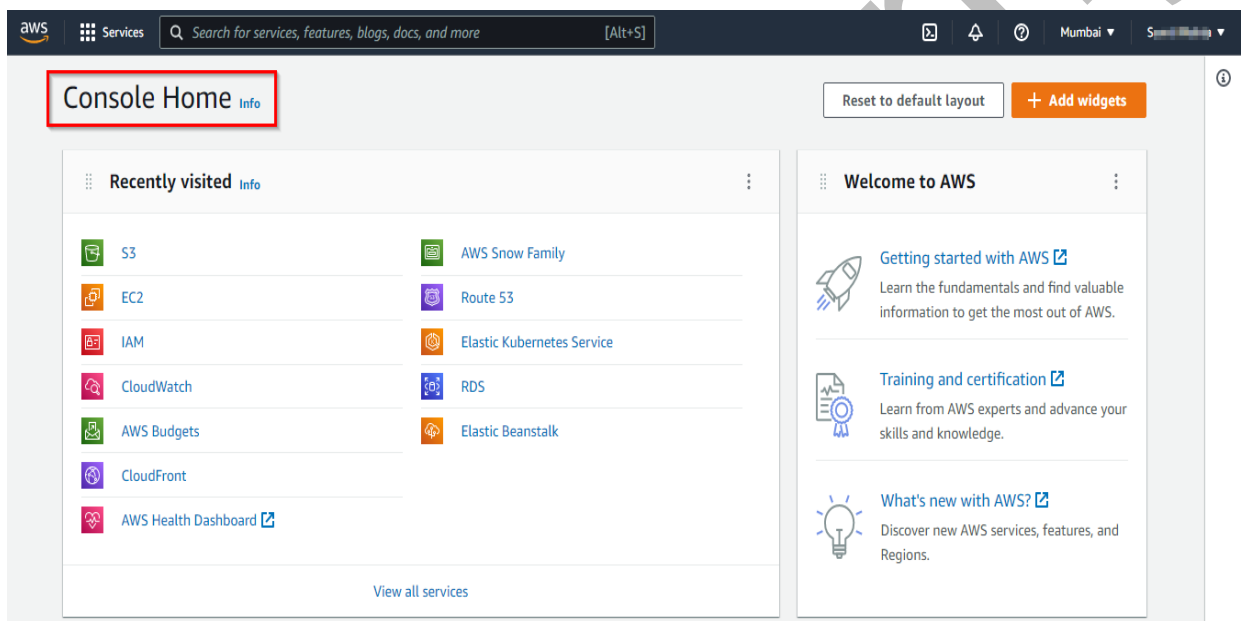
6 AWS IAM POWER USER

Power users are ones who have full access to AWS services but management of IAM service is not allowed to them. IAM power users have broad permissions within an AWS account, allowing them to manage user access and permissions, and control the security and access control of AWS resources.

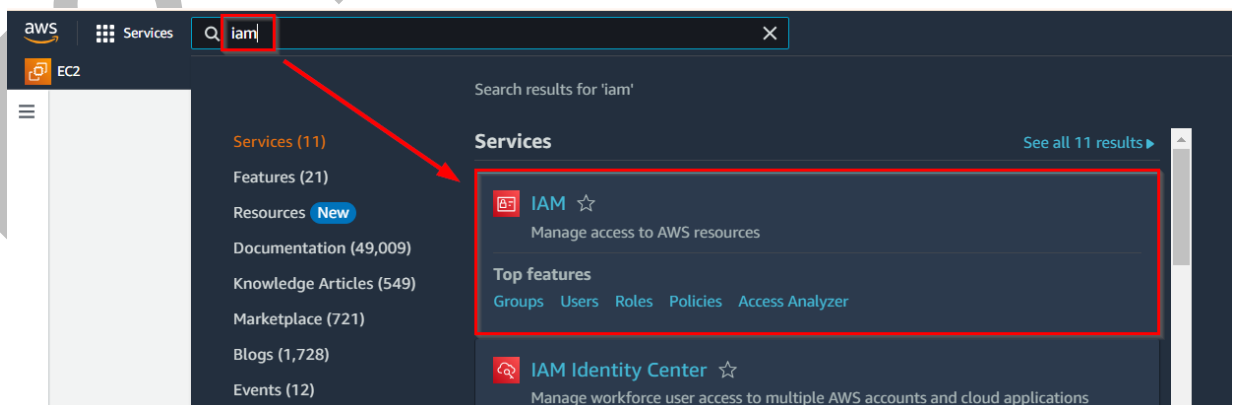
6.1 Create IAM Power User

In this section we are going to create an IAM Power user.

1. Login to you AWS console.



2. Now Search for IAM and Click on it to Open.



3. Click on user from left panel.

The screenshot shows the AWS IAM Dashboard. On the left, the 'Users' link is highlighted in the 'Access management' section. The main dashboard area displays 'Security recommendations' with two items: 'Add MFA for root user' and 'Deactivate or delete access keys for root user'. Each item has a corresponding button ('Add MFA' and 'Manage access keys' respectively).

4. Click on Add user.

The screenshot shows the AWS IAM Users page. The 'Create user' button is highlighted in orange. Below the header, there is a table listing existing users. The table has the following columns: User name, Path, Group, Last activity, MFA, Password age, Console last sign-in, and Access key ID.

User name	Path	Group	Last activity	MFA	Password age	Console last sign-in	Access key ID
Console	/		1:00 days ago	-	1:00 days	August 26, 2023, 1:04...	AKIAI...AKIAH4LPT0S...
Console	/		1:01 days ago	-	1:00 days	October 11, 2023, 11...	-
Telnet	/		1:04 days ago	-	1:00 days	October 30, 2023, 11...	AKIAI...AKIAH4LPT0S...
Server	/		1:11 days ago	-	1:01 days	October 06, 2023, 10...	AKIAI...AKIAH4LPT0S...
Server_K21	/	1	-	-	1:01 days	-	AKIAI...AKIAH4LPT0S...

- Provide the Username, check mark the Enable console access, select create IAM user then set custom password, create the password and Click on **Next**.

IAM > Users > Create user

Step 1 of 4

Specify user details

User details

User name

1

The user name can have up to 64 characters. Valid characters: A-Z, a-z, 0-9, and + = , . @ _ - (hyphen)

2 ☒ Provide user access to the AWS Management Console - *optional*
If you're providing console access to a person, it's a [best practice](#) to manage their access in IAM Identity Center.

3 **Are you providing console access to a person?**

☐ Specify a user in Identity Center - Recommended
We recommend that you use Identity Center to provide console access to a person. With Identity Center, you can centrally manage user access to their AWS accounts and cloud applications.

☒ **I want to create an IAM user**
We recommend that you create IAM users only if you need to enable programmatic access through access keys, service-specific credentials for AWS CodeCommit or Amazon Keyspaces, or a backup credential for emergency account access.

Console password

☐ Autogenerated password
You can view the password after you create the user.

4 ☒ Custom password
Enter a custom password for the user.

- Must be at least 10 characters long
- Must include at least one uppercase letter (A-Z)
- Must include at least one lowercase letter (a-z)
- Must include at least one number (0-9)
- Must include at least one non-alphanumeric character (! @ # \$ % ^ & * () _ + - = [] { } | ')

☐ Show password

5 ☐ Users must create a new password at next sign-in (recommended).

! If you are creating programmatic access through access keys or service-specific credentials for AWS CodeCommit or Amazon Keyspaces, you can generate them after you create this IAM user. [Learn more](#)

Cancel **Next**

- Click on **Attach existing policies directly**, search for **PowerUserAccess** and click on the checkbox and click on Next.

Set permissions

Add user to an existing group or create a new one. Using groups is a best-practice way to manage user's permissions by job functions. [Learn more](#)

Permissions options

☐ Add user to group

Add user to an existing group, or create a new group. We recommend using groups to manage user permissions by job function.

☐ Copy permissions

Copy all group memberships, attached managed policies, and inline policies from an existing user.

☒ **Attach policies directly**

Attach a managed policy directly to a user. As a best practice, we recommend attaching policies to a group instead. Then, add the user to the appropriate group.

Permissions policies (1/1183)

Choose one or more policies to attach to your new user.

Filter by Type
All types
1 match

☒	Policy name	Type	Attached entities
☒	PowerUserAccess	AWS managed - job fun...	0

► Set permissions boundary - optional

Cancel
Previous
Next

- Click on Create User.

Step 1

[Specify user details](#)

Step 2

[Set permissions](#)

Step 3

Review and create

Step 4

[Retrieve password](#)

Review and create

Review your choices. After you create the user, you can view and download the autogenerated password, if enabled.

User details

User name	Console password type	Require password reset
AyushPowerUser	Custom password	No

Permissions summary

Name	Type	Used as
PowerUserAccess	AWS managed - job function	Permissions policy

Tags - optional

Tags are key-value pairs you can add to AWS resources to help identify, organize, or search for resources. Choose any tags you want to associate with this user.

No tags associated with the resource.

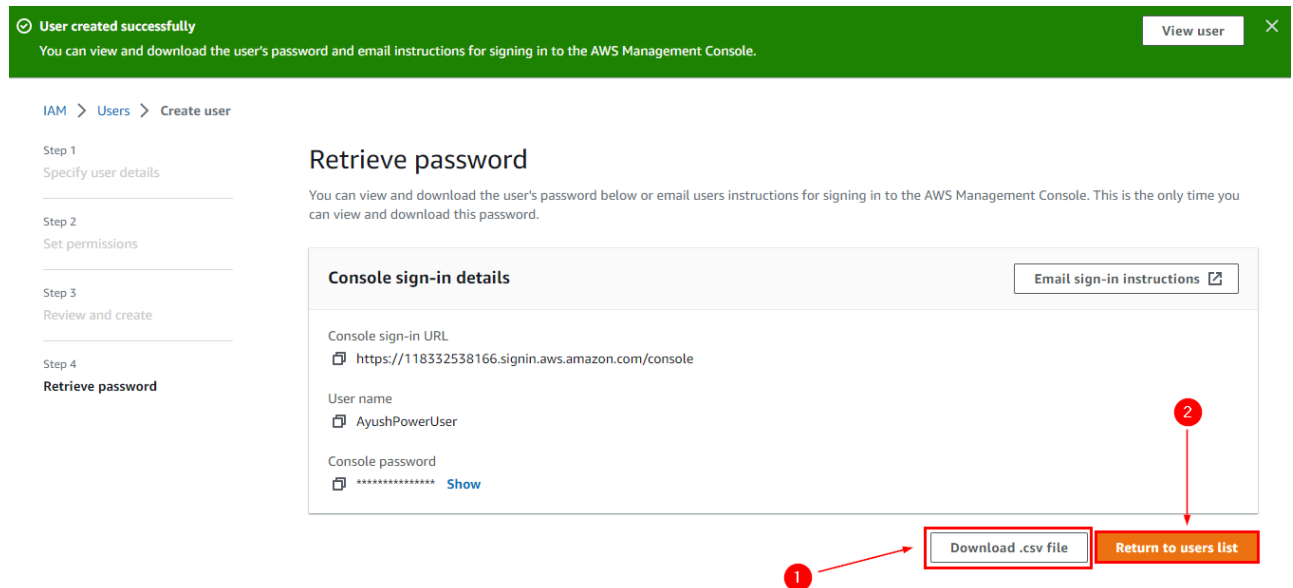
[Add new tag](#)

You can add up to 50 more tags.

Cancel
Previous
Create user

- Click on Download .CSV and click on **Return to users list**.

Note: Here I strongly recommend you download this Csv file from here only, we are not able to download it further.

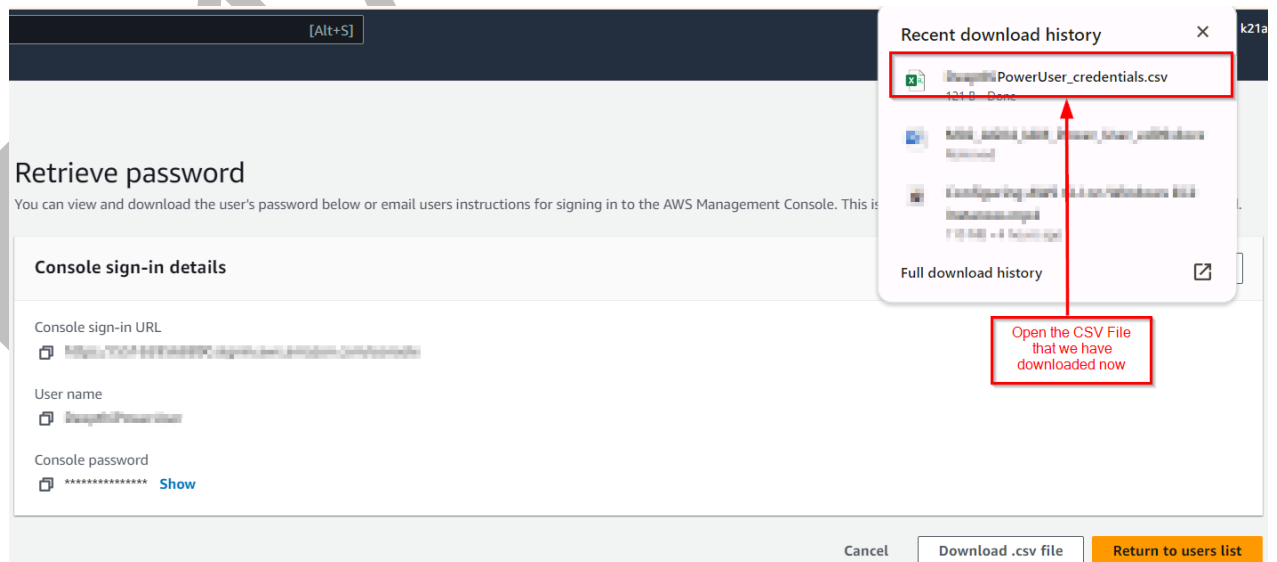


At this step we have successfully created an IAM Power User

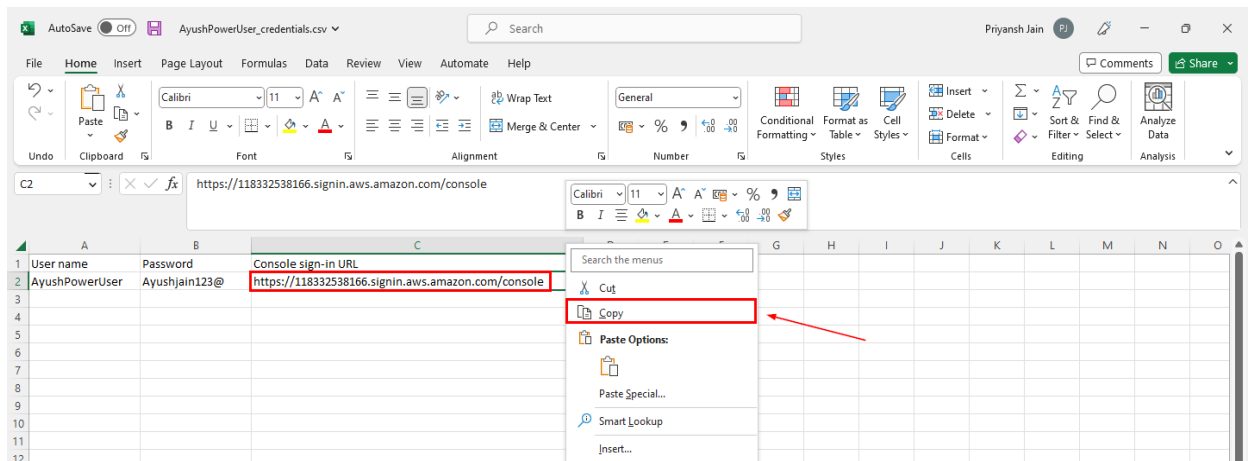
6.2 Login Via IAM Power User

In this section we are going to Login the IAM Power User to the AWS Console.

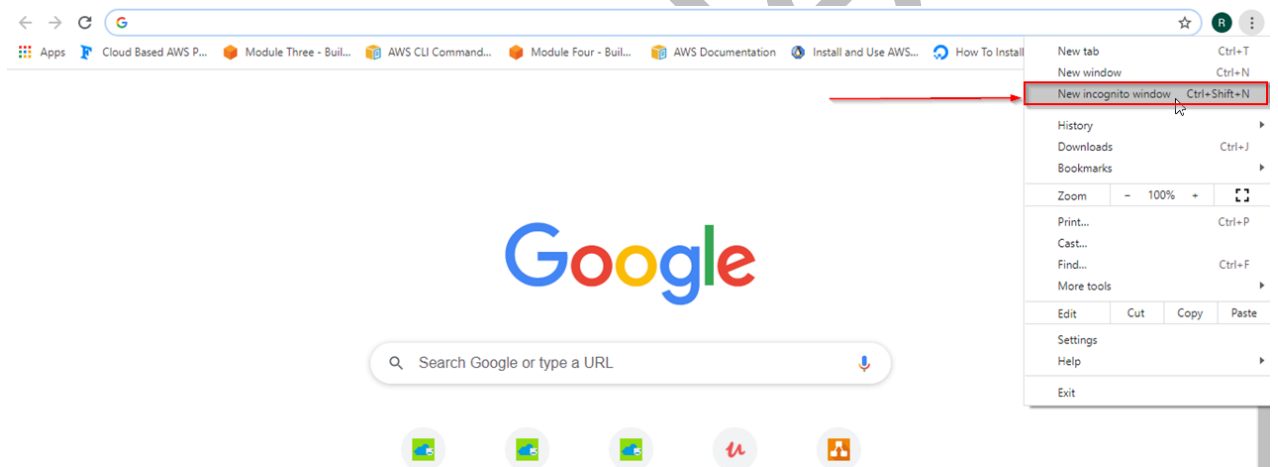
- Open CSV file that we downloaded.



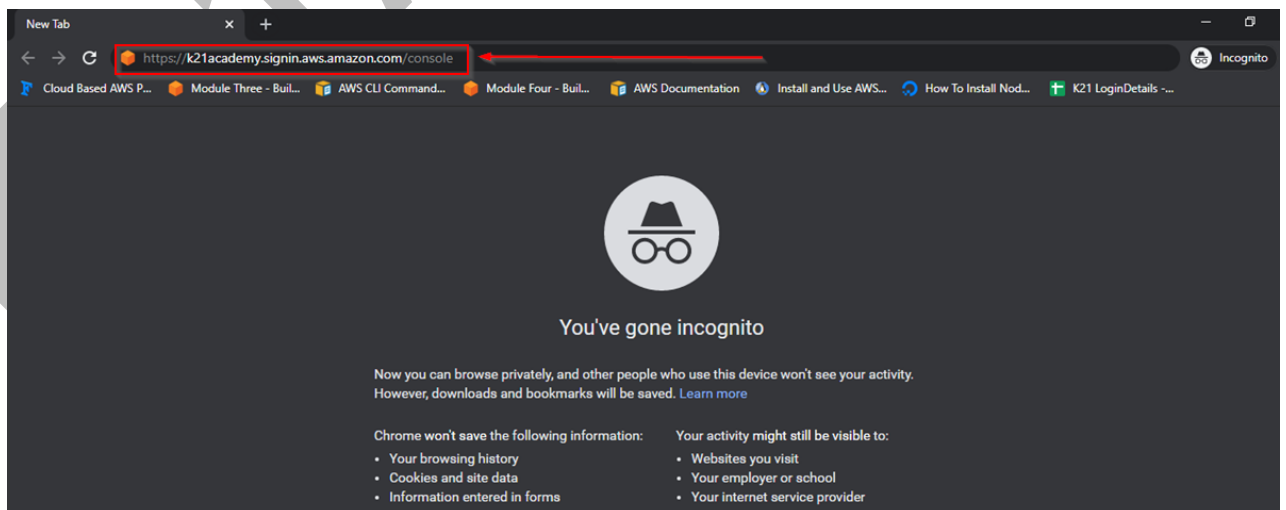
2. Copy the Console Link from the CSV file.



3. Click on the 3 dots and select the New incognito window.



4. Paste the Console Link here in the browser.



5. Provide all the User Credentials and click on Sign in.



Sign in as IAM user

Account ID (12 digits) or account alias

118332538166

IAM user name

AyushPowerUser

Password

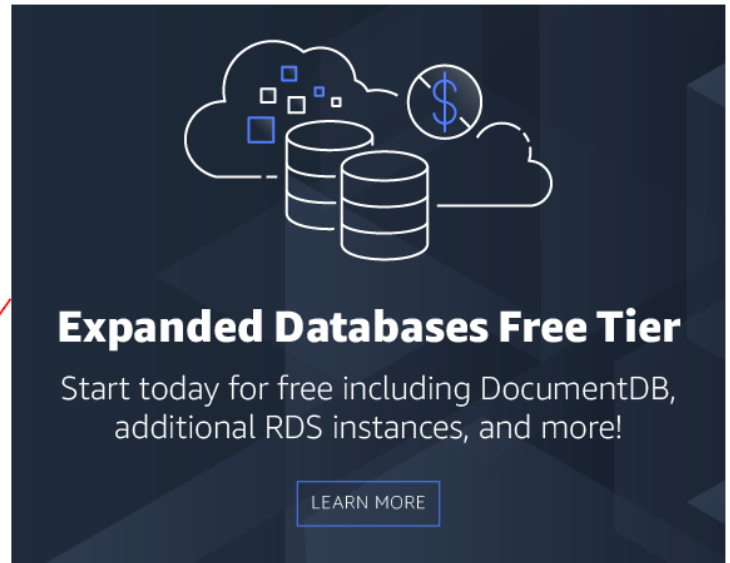
.....

☐ Remember this account

Sign in

[Sign in using root user email](#)

[Forgot password?](#)

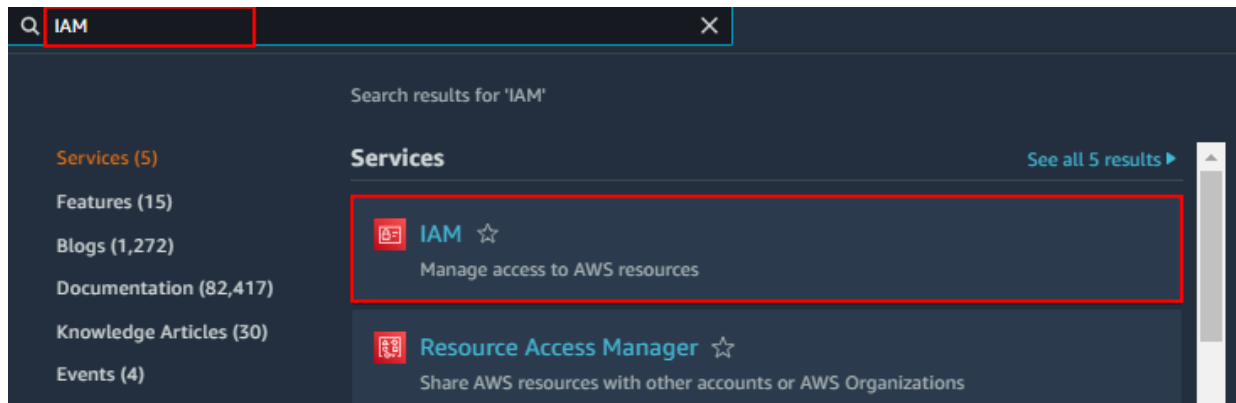


At this step we have successfully logged in with our IAM Power user to the AWS Console

7 TEST ACCESS FOR IAM

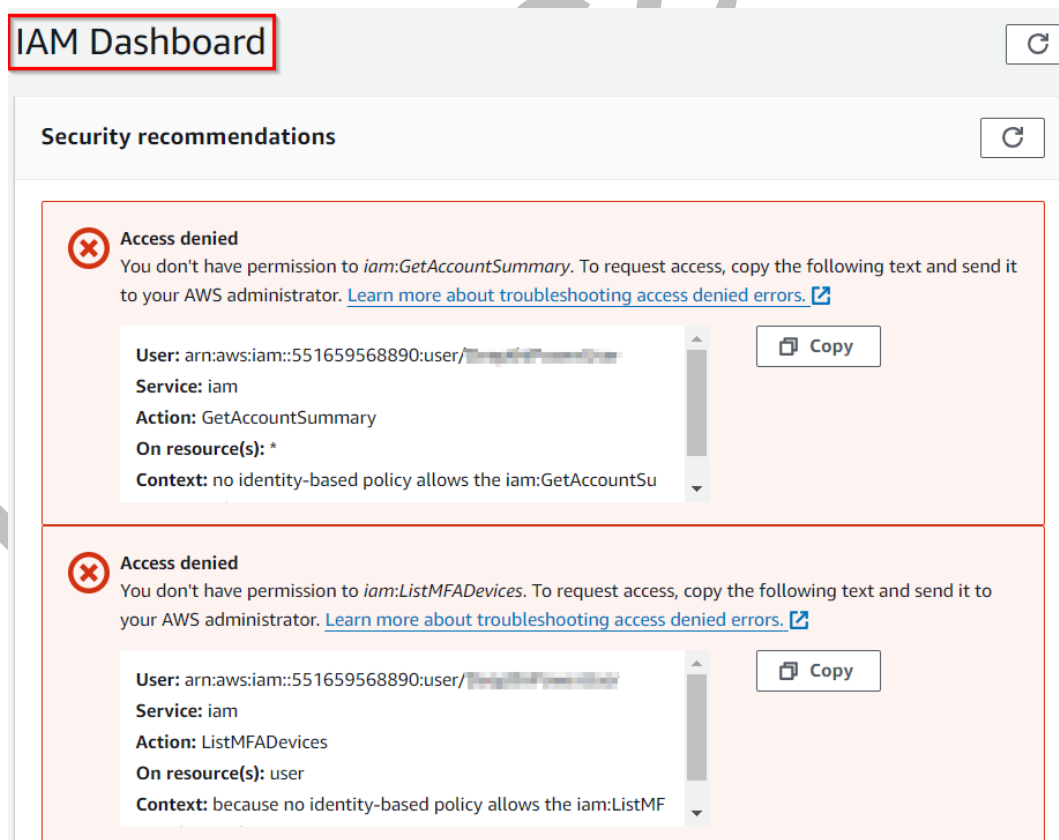
In this section we are going to test whether the Power user has the privilege to access IAM service or not.

1. Open the console and search for IAM



2. Here you see an error message is prompted on your screen while accessing the IAM.

Note: Here the IAM user is not able to access the IAM resources because of its policy that we attached in **Section 4.1 Step 6**.



3. Now click on **User groups** from the left panel.

The screenshot shows the AWS IAM console's 'User groups' page. In the left-hand navigation pane, 'User groups' is highlighted with a red box and a red arrow. The main content area shows a table with columns: Group name, Users, Permissions, and Creation time. Below the table, a red box contains an 'Access denied' message: 'You don't have permission to iam:ListGroup. To request access, copy the following text and send it to your AWS administrator. [Learn more about troubleshooting access denied errors.](#)' The text to be sent is: User: arn:aws:iam::551659568890:user/[redacted], Service: iam, Action: ListGroups, On resource(s): arn:aws:iam::551659568890:group/, Context: no identity-based policy allows the iam:ListGroup action. A 'Copy' button is next to the text.

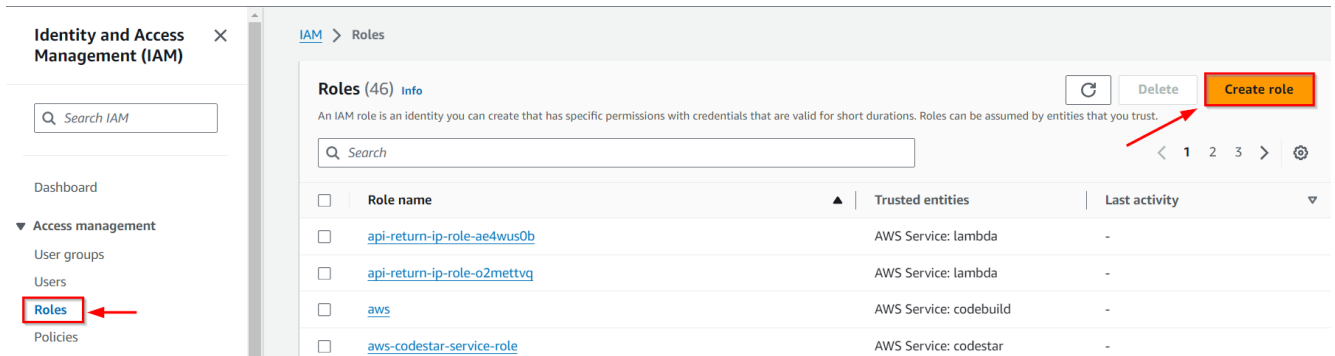
4. Here IAM user is not able to access the groups.

This screenshot is identical to the previous one, showing the 'User groups' page with an 'Access denied' error. The error message and the text to be sent to the administrator are the same: 'You don't have permission to iam:ListGroup. To request access, copy the following text and send it to your AWS administrator. [Learn more about troubleshooting access denied errors.](#)' User: arn:aws:iam::551659568890:user/[redacted], Service: iam, Action: ListGroups, On resource(s): arn:aws:iam::551659568890:group/, Context: no identity-based policy allows the iam:ListGroup action. A 'Copy' button is present.

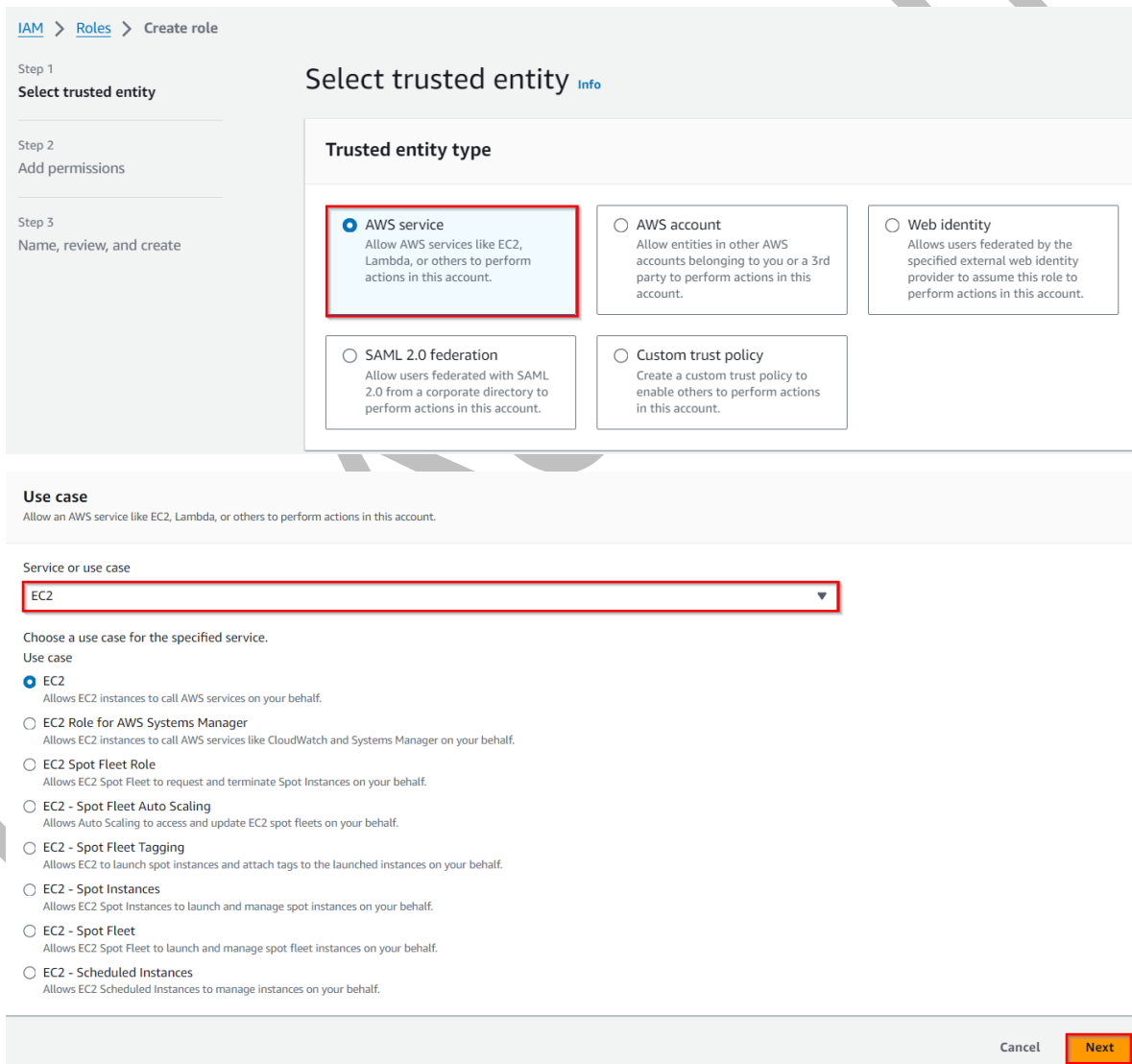
5. While clicking on users, we see that the IAM user is also not authorized for access the users.

The screenshot shows the AWS IAM console's 'Users' page. In the left-hand navigation pane, 'Users' is highlighted with a red box and a red arrow. The main content area shows a table with columns: User name, Path, Group, Last activity, MFA, Password age, Console last sign-in, and Access key ID. Below the table, a red box contains an 'Access denied' message: 'You don't have permission to iam:ListUsers. To request access, copy the following text and send it to your AWS administrator. [Learn more about troubleshooting access denied errors.](#)' The text to be sent is: User: arn:aws:iam::551659568890:user/[redacted], Service: iam, Action: ListUsers, On resource(s): arn:aws:iam::551659568890:user/, Context: no identity-based policy allows the iam:ListUsers action. A 'Copy' button is next to the text.

6. Now click on Roles and try to create a role by clicking on **Create Role**



7. Now click on **AWS Services** and select any use case and click on **Next Permission**.



Step 1
Select trusted entity

Step 2
Add permissions

Step 3
Name, review, and create

Select trusted entity

Trusted entity type

- ☒ **AWS service**
Allow AWS services like EC2, Lambda, or others to perform actions in this account.
- ☐ **AWS account**
Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.
- ☐ **Web identity**
Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.
- ☐ **SAML 2.0 federation**
Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.
- ☐ **Custom trust policy**
Create a custom trust policy to enable others to perform actions in this account.

Use case
Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

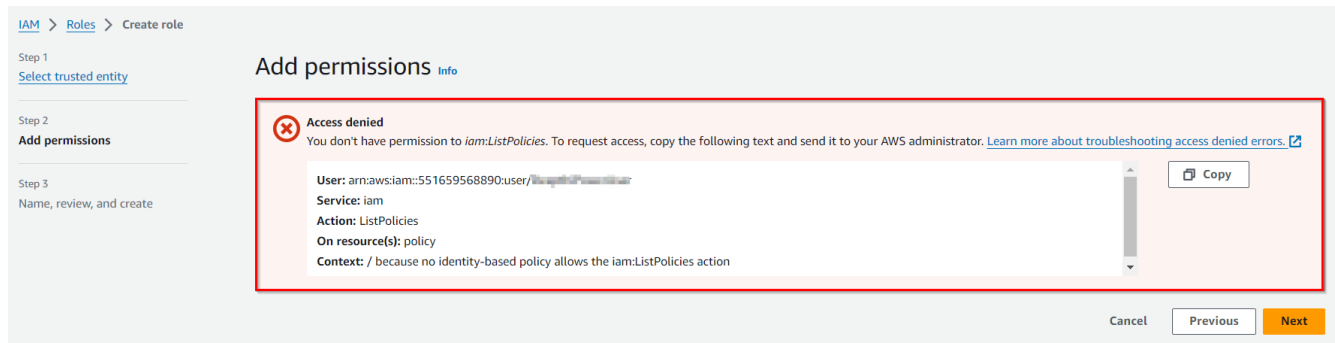
Service or use case
EC2

Choose a use case for the specified service.
Use case

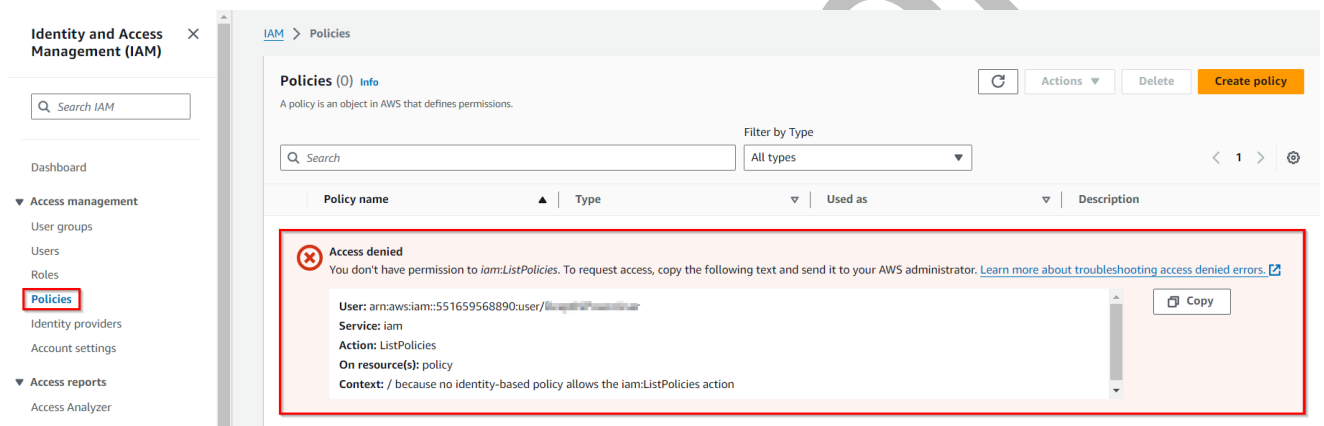
- ☒ **EC2**
Allows EC2 instances to call AWS services on your behalf.
- ☐ **EC2 Role for AWS Systems Manager**
Allows EC2 instances to call AWS services like CloudWatch and Systems Manager on your behalf.
- ☐ **EC2 Spot Fleet Role**
Allows EC2 Spot Fleet to request and terminate Spot Instances on your behalf.
- ☐ **EC2 - Spot Fleet Auto Scaling**
Allows Auto Scaling to access and update EC2 spot fleets on your behalf.
- ☐ **EC2 - Spot Fleet Tagging**
Allows EC2 to launch spot instances and attach tags to the launched instances on your behalf.
- ☐ **EC2 - Spot Instances**
Allows EC2 Spot Instances to launch and manage spot instances on your behalf.
- ☐ **EC2 - Spot Fleet**
Allows EC2 Spot Fleet to launch and manage spot fleet instances on your behalf.
- ☐ **EC2 - Scheduled Instances**
Allows EC2 Scheduled Instances to manage instances on your behalf.

Cancel **Next**

8. Here you again see the error message is prompted on your screen while creating a **Role**



9. Now while clicking on policies, here you again see an error message on your screen regarding your permission.



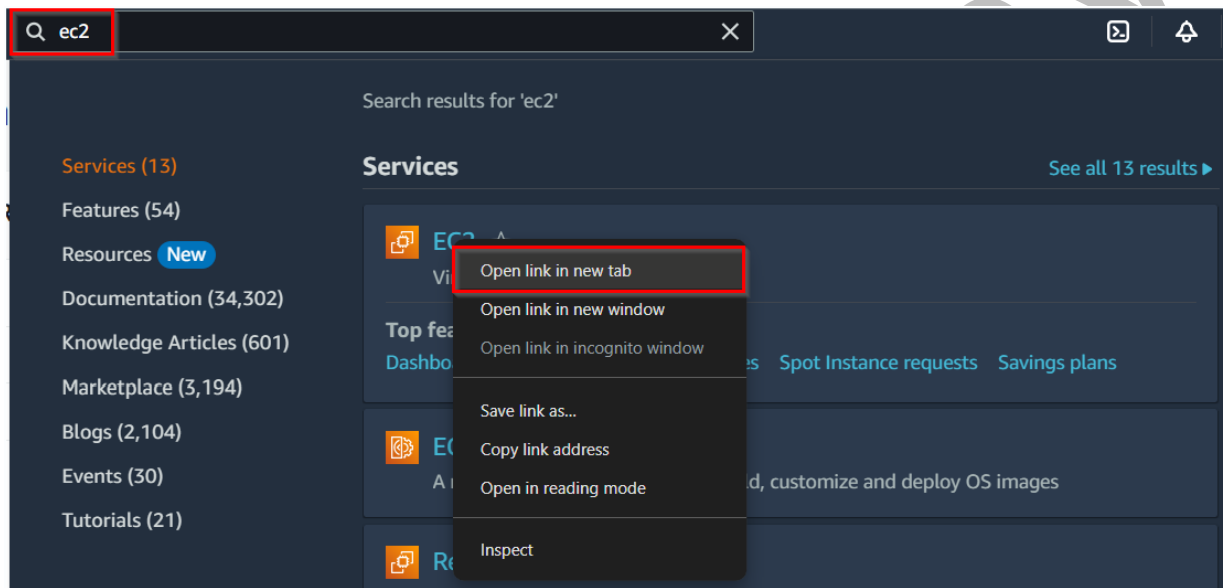
8 TEST OTHER AWS SERVICE ACCESS

In this section we are going to test whether the Power user have the privilege to access other AWS services or not. We will cover these AWS services in further modules.

8.1 AWS EC2 Service

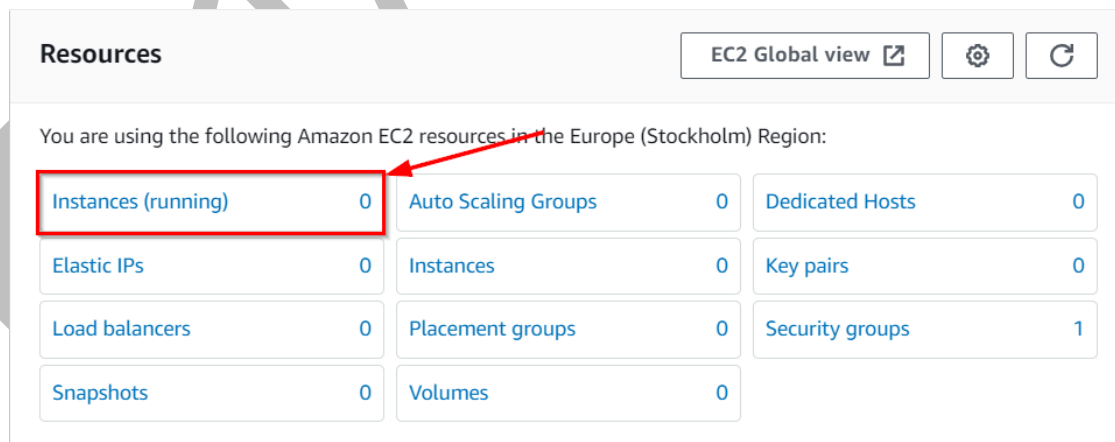
In this section we are going to test the EC2 service is accessible by IAM Power user or not.

1. Search for EC2 and Open in new tab



2. Here you shall see the EC2 dashboard.

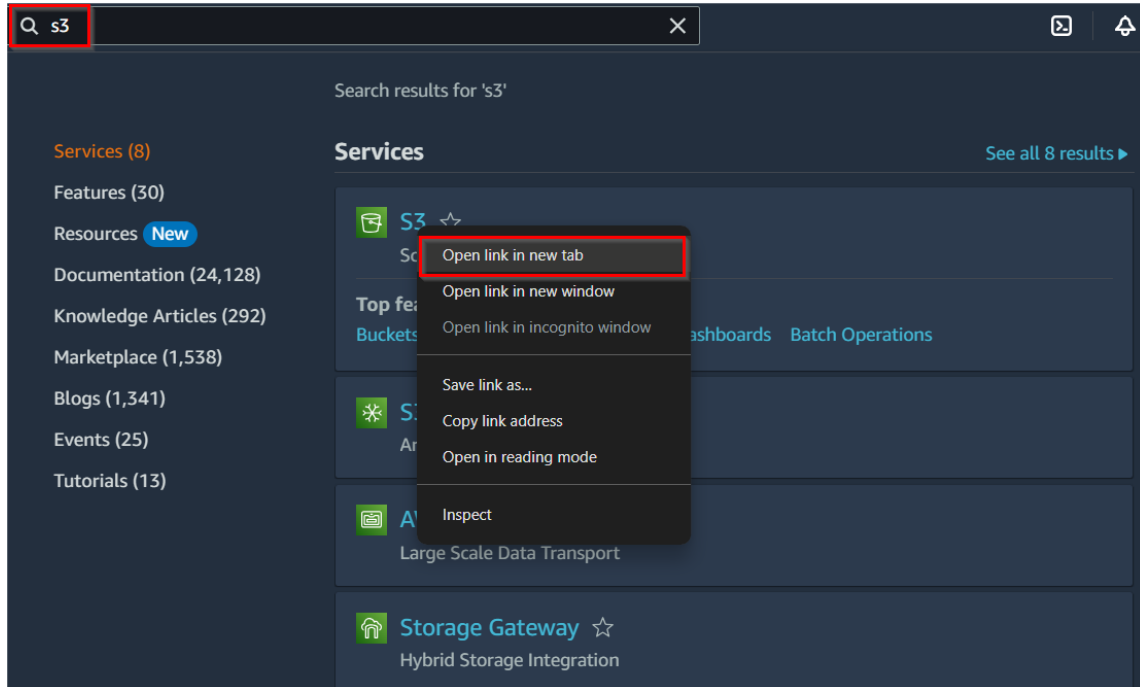
Note: Here we are able to access the EC2 services



8.2 AWS S3 Service

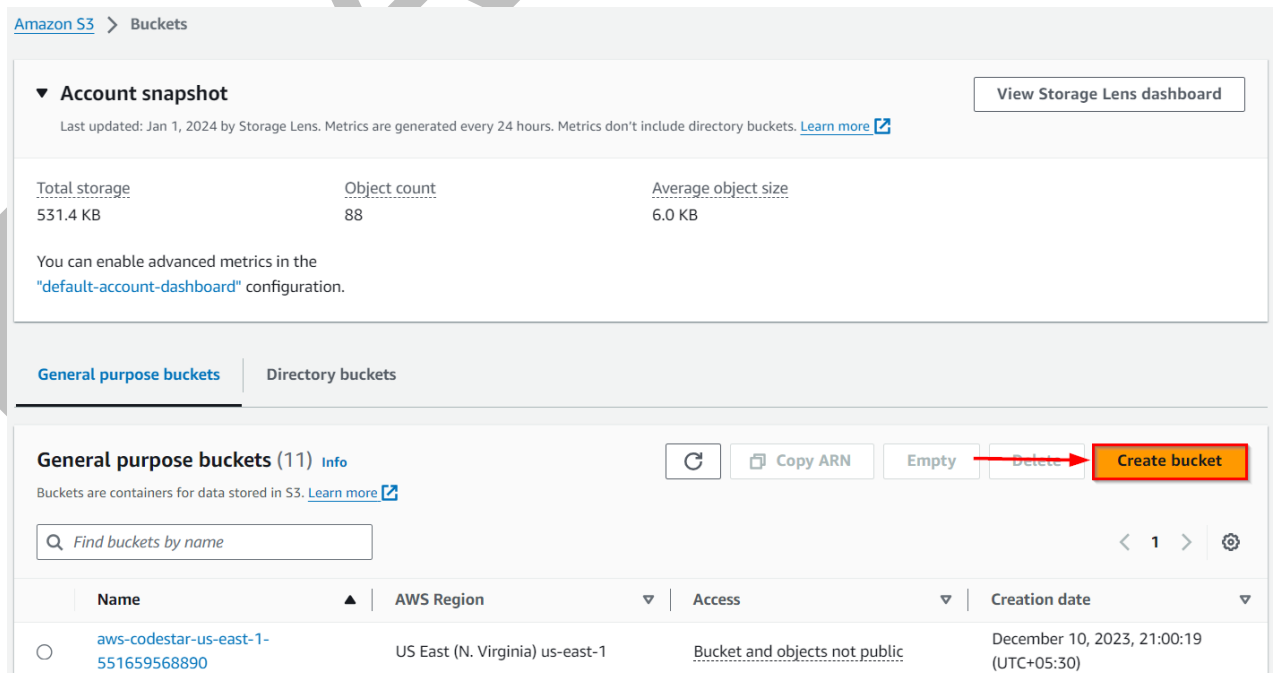
In this section we are going to test the S3 service is accessible by IAM Power user or not.

1. Search for S3 and Open in new tab



2. Here you shall see the AWS S3 dashboard.

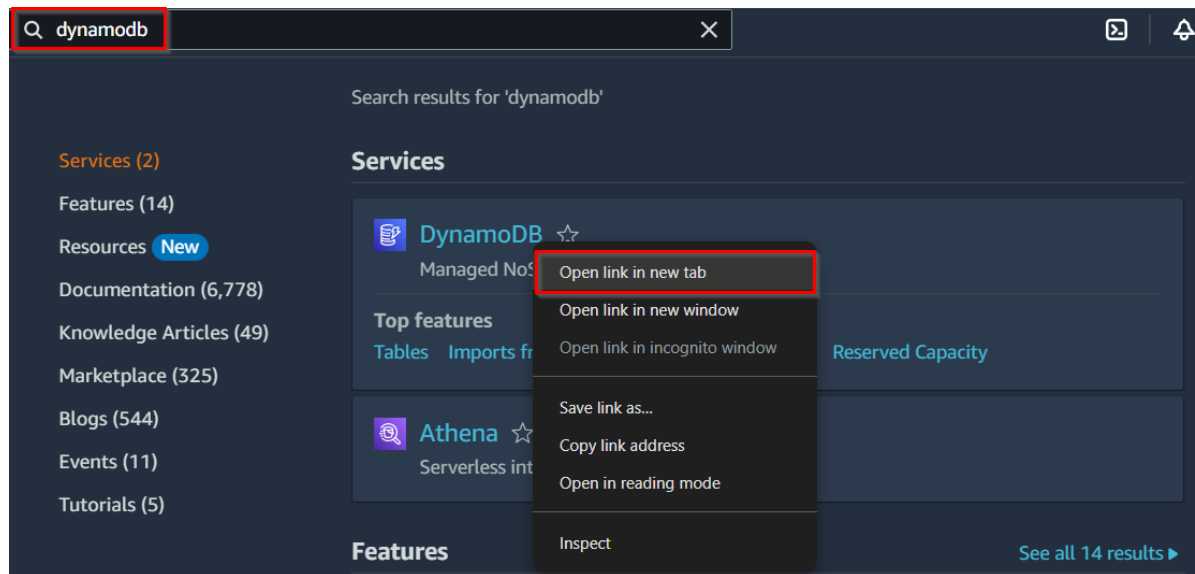
Note: Here we are able to access the AWS S3 also



8.3 AWS DynamoDB

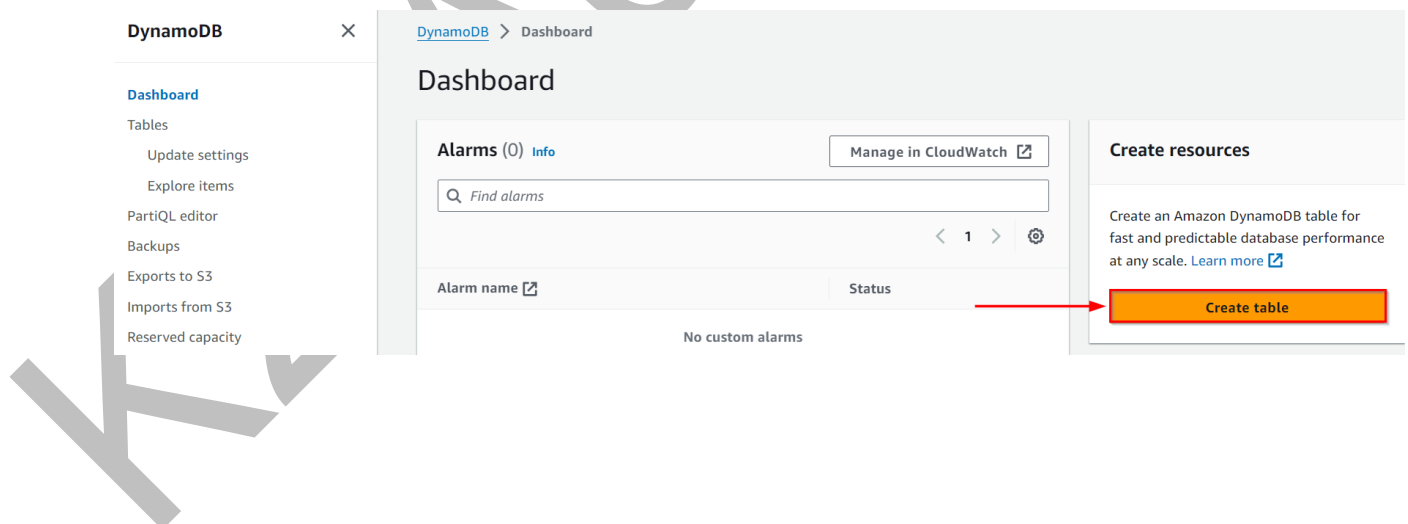
In this section we are going to test the DynamoDB service is accessible by IAM Power user or not.

1. Search for DynamoDB and Open it in New Tab



2. Here you shall see the DynamoDB Dashboard

Note: Here we are able to access the DynamoDB service.

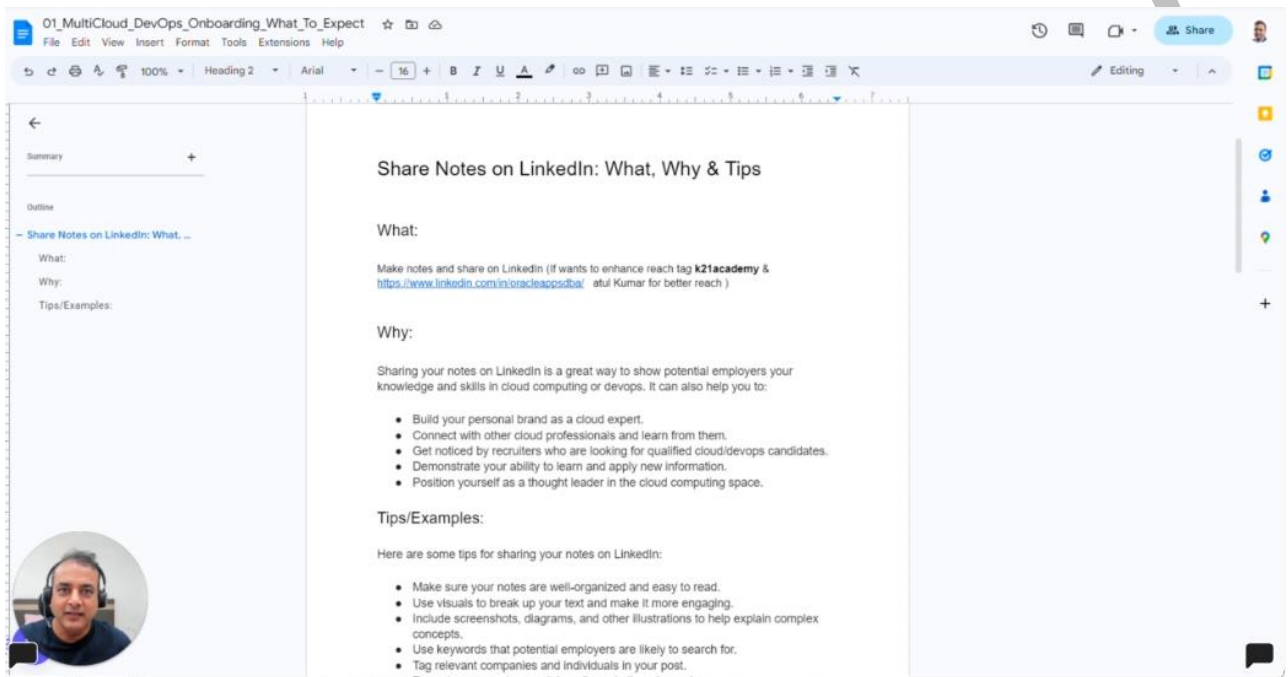


9 SHARE YOUR LEARNINGS

In this section, you are going to understand how to share whatever you learned during this lab on LinkedIn and WhatsApp.

Please Watch the below 🖱️ video to understand how you can achieve this:

🖱️ <https://www.loom.com/share/d91f9abbec794f4f85e7d99bb82c2ff6>



🖱️ As you know, building a strong professional profile is essential in today's job market. One way to showcase your skills and knowledge is by sharing your labs and projects on LinkedIn.

Your LinkedIn profile is a powerful tool in your job search. Sharing your labs and projects is just one way to demonstrate your expertise and stand out to potential employers. If you don't have LinkedIn, we strongly recommend you create one for yourself.

🖱️ By doing so, you can demonstrate to potential employers or connections that you have hands-on experience in your field and are actively engaged in learning and growing your skills.

9.1 On LinkedIn

Take a screenshot of lab and share it on your LinkedIn. So, this will attract recruiters and employers on your profile and increase reach. Do remember to tag **K21Academy**, as we will circulate in our network too, to increase your reach.

Please refer these following examples on how to share on LinkedIn (don't limit yourself to just these type). Share learnings in points, notes 📝 or anything that can help others.

Here is a sample which includes steps inside a post followed by screenshot:

Sumti Mehta • 1st
Cloud Computing | Amazon Web Services | 1x AWS Certified | Technical Support...
2w • Edited • 🌐

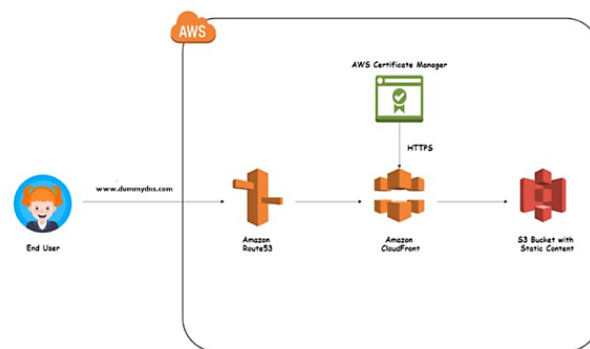
Hi There,

Today I have Provisioned an SSL/TLS Certificates using AWS Certificate Manager. It was fun and learned a lot of services integrated in the deployment.

To do this deployment I had to gone through the following steps:

1. Created an S3 bucket and enabled Static Website Hosting
2. Created a Hosted Zone and Mapped DNS Using Route53
3. Requested SSL/TLS certificates using AWS Certificate Manager
4. And finally, Configured CloudFront Distribution

Thanks, [K21Academy: Learn Cloud From Experts](#) For providing me with the required resources. [#aws](#) [#awscloud](#) [#awsservices](#) [#learning](#) [#cloud](#) [#cloud](#) [#project](#) [#learning](#)



Sumti Mehta and 15 others

Sahid Anwar Hussain • 1st
DevOps Engineer & Cloud Support at K21Academy: Learn Cloud From Experts
2w • 🌐

Hi There,

Today I have deployed an awesome tool to monitor an entire K8s cluster. It was fun and learned a lot diring the deployment.

To do this deployment I had to gone through the following steps:

1. Created an multinode K8s cluster.
2. Created a namespace named "monitoring".
3. Created clusterRole & clusterRole Binding for Prometheus.
4. Created the ConfigMap encluing all the necessary Data for deploying Prometheus
5. Created a Deployment file for Prometheus
6. Finally I ran all the files and wallah.. Its done.

Thanks, [K21Academy: Learn Cloud From Experts](#) For providing me with the required resources. [#kubernetes](#) [#k8s](#) [#devops](#) [#learning](#) [#cloud](#) [#cloud](#) [#project](#) [#learning](#) [#prometheus](#) [#monitoring](#)

The screenshot shows the Prometheus Targets page. It displays a table of monitoring targets with columns for Endpoint, State, Labels, Last Scrape, and Scrape Duration. The targets are categorized by namespace: kube-state-metrics (0/1 up), kubernetes-apisservers (1/1 up), and kubernetes-cadvisor (3/3 up). The kube-state-metrics target is currently down, while the others are up.

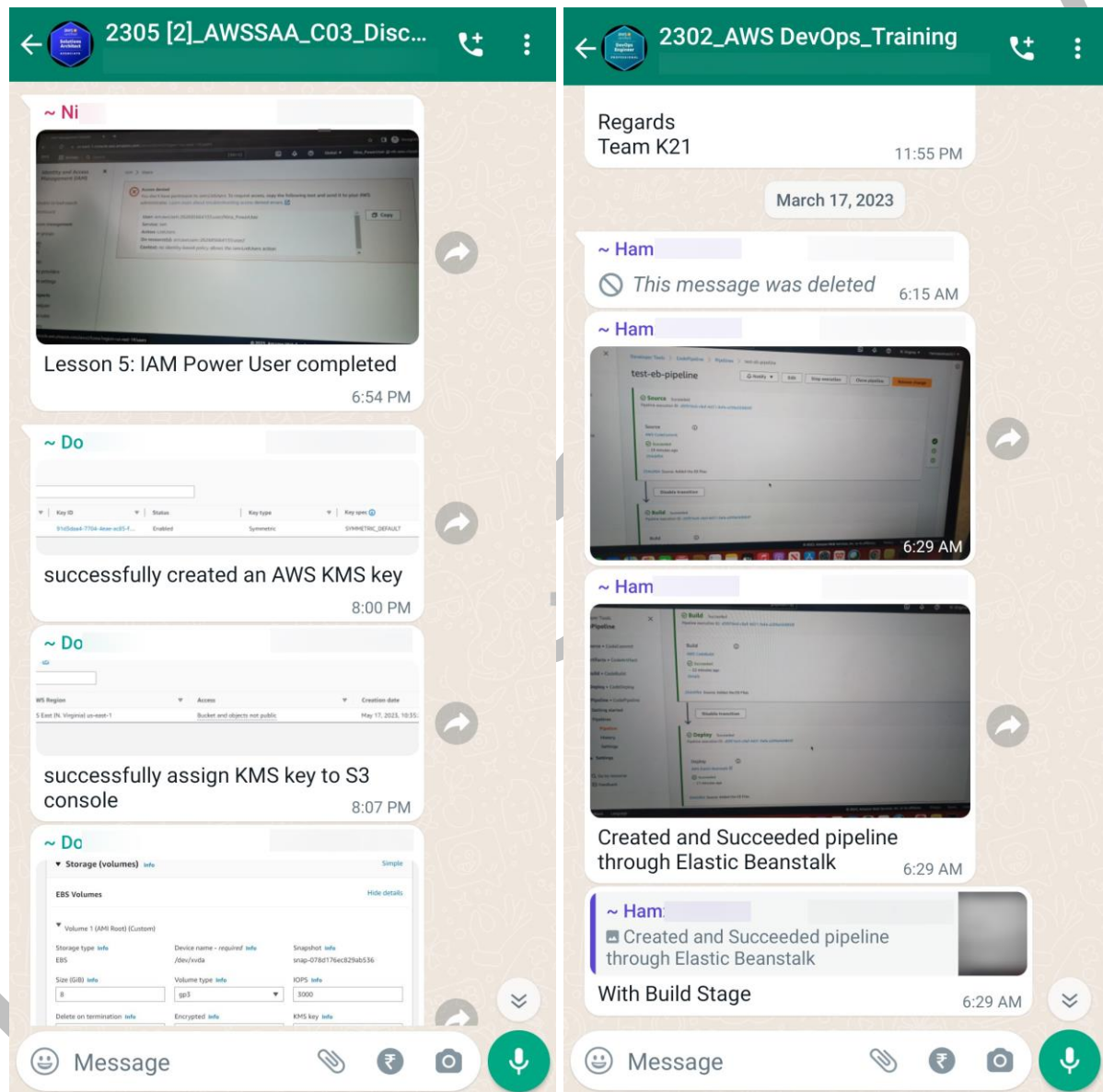
Endpoint	State	Labels	Last Scrape	Scrape Duration	Error
http://kubernetes-master:8080/metrics	DOWN	kubernetes-master	3.990s ago	12.422ms	Get "http://kubernetes-master:8080/metrics": dial tcp: lookup kubernetes-master: no such host
https://70.3.0.4443/metrics	UP	kubernetes-apiservers	1.931s ago	95.479ms	
https://kubernetes-master:8080/metrics	UP	kubernetes-cadvisor	3.493s ago	36.210ms	

Suraj . and 12 others

9.2 On WhatsApp

Share screenshot of lab in WhatsApp group so this will encourage others and also boost your confidence on the progressive path you are following.

Also, we will keep track of your progress on labs so do share your screenshot in WhatsApp. group too.



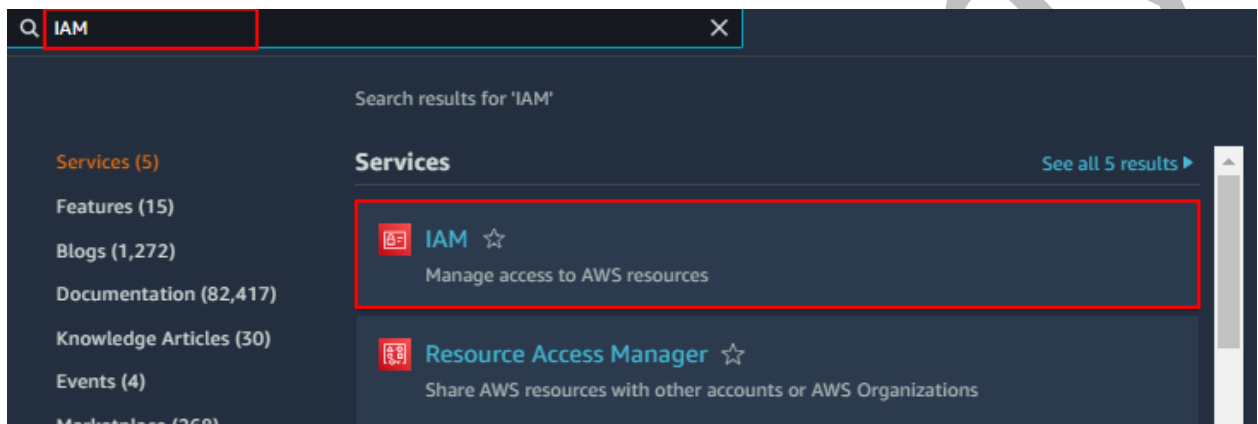
10 DELETE AND CLEAN-UP

There is No charge for creating a user but keeping the resources running you shall be charged. So, to complete all the labs under free tier, please delete all the resources that are no longer required.

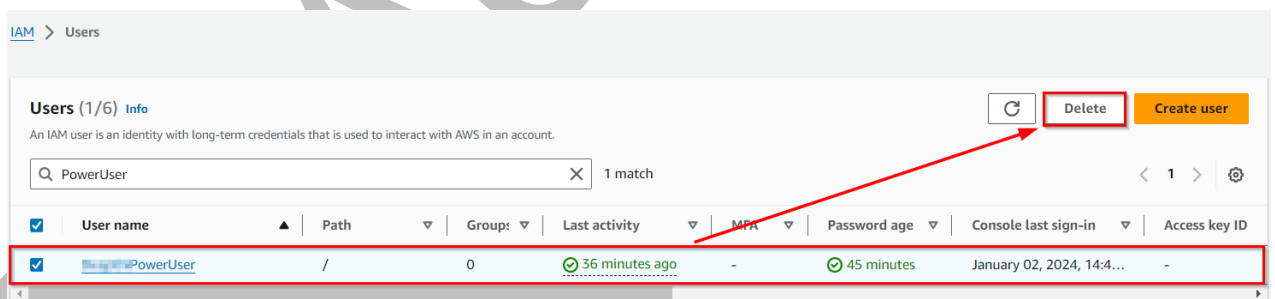
10.1 Delete IAM Power User

Here we are going to delete the IAM power user which we have created earlier.

1. Back to your root account and search for IAM and open it.



2. Click on **users** from the left panel, select the user and click on **Delete**.



- Confirmation pop-up is prompted on the screen, type the **Username** and click on **Delete** button.

Delete **PowerUser**? ×

Delete **PowerUser** permanently? This will also delete all its user data, security credentials and inline policies.

User name	Last activity
PowerUser	39 minutes ago

Note: Recent activity usually appears within 4 hours. Data is stored for a maximum of 365 days, depending when your region began supporting this feature. [Learn more](#)

This action cannot be undone.

To confirm deletion, enter the user name in the text input field.

PowerUser

Cancel

Delete user

At this step we have successfully deleted our IAM Power User

11 TROUBLESHOOTING

Note: In this section, we will document any issues and their fixes that may arise while performing this lab. If you encounter any issues, please let us know so that we can update this section with the relevant information and help others who may face similar problems.

K21Academy

12 SUMMARY

This activity guide cover steps for:

1. Creating a IAM Power User
2. Login the IAM Power user to AWS Console
3. Test IAM service Access
4. Test other service Access